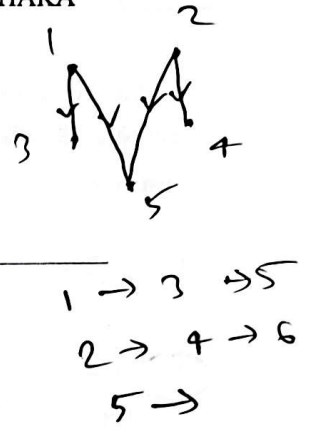


SECTION – AThere are **FOUR** questions in this section.Answer **Question No. 1 (compulsory)** and any other **TWO** from the rest.

1. (a) Given an adjacency list representation of a directed Graph, $G = (V, E)$, sketch an efficient algorithm to find its transpose graph $G^T = (V, E^T)$, where $E^T = \{(v, u) \in V \times V : (u, v) \in E\}$. Analyze the time complexity of your algorithm. Also, evaluate how long you take to compute the in-degree and out-degree of every vertex of G . (18)

- (b) State the correctness theorem of Breadth-First Search (BFS). Use this theorem to judge whether the value $u.d$ assigned to a vertex u in BFS depends on the order of appearance of the vertices in the corresponding adjacency lists. Now demonstrate with the example of the graph given in Fig. for Q. No. 1(b) whether the breadth-first tree produced by BFS depends on the ordering of vertices in the adjacency lists if the same source vertex is used in every ordering. (17)

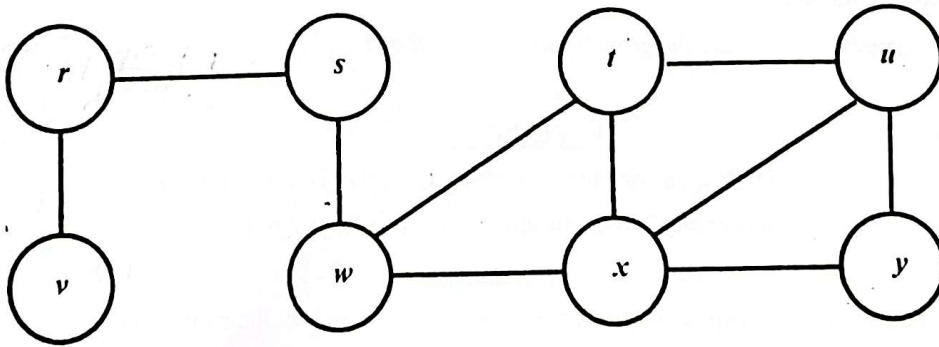


Fig. for Q. No. 1(b)

2. (a) Define the four types of edges that a depth-first search (DFS) on a directed graph yields. Modify the pseudocode of the original DFS algorithm to print all the edges of a directed graph along with their types. Justify whether your algorithm will work correctly. (18)
- (b) Given two max-priority queues represented by two arrays, write an efficient algorithm to merge them into a single max-priority queue. Analyze the time complexity of your algorithm. (17)

CSE 105

3. (a) The key at index 6 of the following max priority queue has been increased to 32. Re-establish the max-heap property of the queue and illustrate the steps using array only.

Justify whether you can use max-heap algorithm here to re-establish heap property. (12)

1	2	3	4	5	6	7	8	9	10	11	12	13
25	23	20	15	18	16	12	14	7	8	11	4	14

32

- (b) Implement a Circular Queue using an array in Java. Include the following methods: (15)

- enqueue(x) - Adds an element x to the queue.
- dequeue() - Removes and returns the front element from the queue.
- is_empty() - Checks if the queue is empty.
- is_full() - Checks if the queue is full.
- front() - Returns the front element without removing it.

- (c) Write a function to delete a node with a given key from a doubly linked list. Assume the list contains unique keys and you have access to both the head and tail of the list. (8)

4. (a) How would you reverse a singly linked list (without duplicating the list)? Write the steps (pseudocode) involved conceptually. (10)

- (b) How can you remove duplicate values from a sorted singly linked list? (5)

- (c) How can you check if a singly linked list is a palindrome? Write the steps involved conceptually. (10)

- (d) Analyze the compare the growth rate of $n^{\log \log n}$ and $\log^n n$. (10)

SECTION - B

There are FOUR questions in this section. Question No. 5 is mandatory.

Answer any TWO questions from questions No. 6-8.

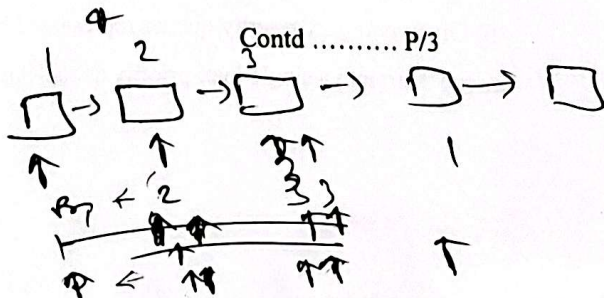
5. (a) You are given a sequence of matrices $\langle A_1, A_2, \dots, A_n \rangle$ and their dimensions $\langle p_0, p_1, p_2, \dots, p_n \rangle$ such that matrix A_i has dimensions $p_{i-1} \times p_i$. The matrix-chain multiplication (MCM) problem aims to identify the minimum number of scalar multiplications necessary to determine the product of the given chain of matrices. (8+7+10=25)

- (i) Formulate arguments to show that the MCM problem has optimal substructure property.

- (ii) Compose a recursion tree diagram to show that MCM has overlapping subproblem property.

- (iii) Design an efficient algorithm using dynamic programming approach to solve this problem.

4-8 3



CSE 105

Contd ... Q. No. 5

(b) Using a dynamic programming algorithm, determine the minimum cost alignment of the two strings X = ATCGGT and Y = ACCDT where both mismatch penalty and gap penalty are 1 and the cost of a match is 0. You do not need to write the algorithm, just show the dynamic programming (DP) table. Show the alignments and mark the paths (which correspond to the alignments), in the DP table. (10)

6. (a) **Explain** the run-time complexity of the merge sort algorithm. In your explanation, first, formulate the run-time of the merge-sort algorithm as a recurrence relation. Then solve the recurrence relation using both (2+5+3=10)

(i) Recurrence tree method, and

(ii) Master theorem (distinguish the case that is applicable in your analysis).

(b) You are asked to sort the array of integers given below in ascending order. Use the Quicksort algorithm to sort the array. Highlight the pivot item at each step and show the status of the array after processing each pivot element. (15)

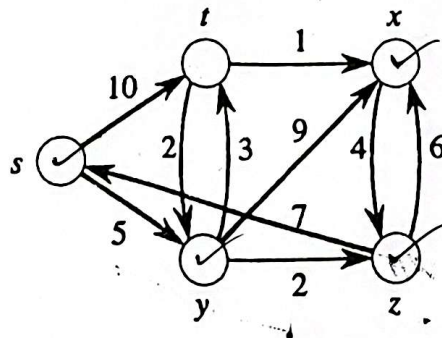
[5, 17, 2, 4, 16, 8, 10, 6, 4, 26, 20, 7]

(c) Construct a proof to show that any comparison-based sorting algorithm requires $O(n \log n)$ comparisons in the worst case. (10)

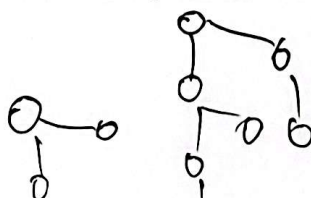
$P \{ \} \leq n$

7. (a) **Argue** in favor of the correctness of the Prim's algorithm to find a minimum spanning tree of a graph $G = (V, E)$. Prove any non-trivial graph properties used in the argument. (5+10=15)

(b) **Find** the shortest path distance from node x to node t in the following graph using the Dijkstra's Algorithm. Re-draw the diagram for each intermediate step of the algorithm (after relaxation) while annotating each node with appropriate shortest path distance estimate of that step. (10)



(c) Consider the array implementation of the Union-Find data structure for some set S of size n, where unions keep the name of the larger set. **Analyze** the run-time complexity of any sequence of k Union operations. (10)



Contd P/4

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CSE 105

8. (a) Prove the following theorems that describe the properties of perfect binary trees: **(4+6=10)**

(i) A perfect tree has $2^{h+1} - 1$ nodes

(ii) The average depth of a node is $\Theta(\ln(n))$

(b) Write a code snippet to traverse the binary tree in an order so that you can store it in an array. Show the placement of numbers of the following tree in an array. Write a code to access the parent of a given node. What will be the index no of the array for node 12? **(10+3+8+4=25)**

